

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

## Practical - VIII

|                              |                                   |
|------------------------------|-----------------------------------|
| Roll No. :S030               | Name:Pradeep Rajbhar              |
| Class :SYIT                  | Batch : 01                        |
| Date of Assignment :12/08/26 | Date/Time of Submission :12/08/26 |

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

- a. Bubble sort

**Code:**

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def bubblesort(array):
    for i in range(len(array)):
        for j in range(0, len(array)-i-1):
            if array[j]>array[j+1]:
                temp =array[j]
                array[j]=array[j+1]
                array[j+1]=temp

data=[-2,4,0,11,-9,12,99,-25]
bubblesort(data)

print('sorted Array in Ascending Order:')
print(data)
|
```

## Output:

```
=====
sorted Array in Ascending Order:
[-25, -9, -2, 0, 4, 11, 12, 99]
>>>
```

b. Insertion sort

## Code:

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def insertionSort(array):
    for step in range(1,len(array)):
        key = array[step]
        j = step - 1

        while j >=0 and key < array[j]:
            array[j+1]=array[j]
            j= j-1

        array[j+1] = key

data = [9,12,8,6,4,3,0]
insertionSort(data)
print("Sorted Array in Ascending Order:")
print(data)
```

## Output:

```
=====
Sorted Array in Ascending Order:
[0, 3, 4, 6, 8, 9, 12]
>>>
```

### c. Selection sort

#### Code:

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] < array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, 6, 12, 22, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)
```

#### Output:

```
==== RESTART: C:/Users/itlab/Desktop/
Sorted Array in Ascending Order:
[-9, -2, 0, 6, 11, 12, 22, 45]
>>>
```

#### Curiosity Questions:

#### Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.